



## Research papers

# Preferential water flow: Influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration

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## ABSTRACT

Soil water infiltration is an important part of the land surface hydrological cycle, and plays an important role in the hydrological response of the soil, such as soil erosion. High infiltration rates favor an increase in the soil water storage capacity that allows maintaining vegetation restoration in arid and semi-arid regions. Alfalfa (*Medicago sativa* L.) is a quality perennial legume grassland, which is widely planted in semi-arid areas. In this study, the effects of the root channels formed by the decay of alfalfa on preferential flow were evaluated as a driving force to improve soil water infiltration and soil water supply. A double-ring infiltrometer (30-cm inner diameter and 60-cm outer diameter) was used to measure the infiltration process with the falling head method. Methylene blue was used to visualize the pathways followed by the infiltrated water. The results showed that the initial infiltration rate in the alfalfa grassland increased by 27.7%, compared with the control bare land, and the total cumulative infiltration was 1.13 times higher. The steady infiltration rate of the grassland increased by 31.8% compared to the bare land. The root channel diameters were measured with a Vernier caliper, and the average root channel area (RCA) was calculated through the average root channel diameter (ARCD). The values of ARCD and RCA were significantly and positively correlated with the infiltration rates, being the coefficients of determination 0.815 and 0.789, respectively. Our results indicated that root channels formed by the decayed roots of alfalfa played an important role in increasing soil water infiltration and soil water supply under semi-arid conditions. Our research improves the understanding of the hydrological cycle processes at the plant-soil interface in semi-arid areas.

## 1. Introduction

Land surface water resources shortage and their effective utilization are among the most serious problems in farmlands, particularly in arid and semi-arid regions (Yin et al., 2018). Soil water is one of the most important limiting factors that influences vegetation growth and distribution patterns in arid and semi-arid areas (Wang et al., 2007, 2008; Wu et al., 2017). Under these conditions, any decrease in soil water could aggravate soil water shortage in both the top and deep soil layers, and even favor the formation of the dry soil layer, further endangering the sustainability of vegetation restoration (Jia et al., 2017, 2019). Furthermore, soil water availability for plants is a clear limiting factor that affects vegetation survival, growth and restoration in arid and

semi-arid areas, because rainfall is the only source that can replenish soil water (Wang et al., 2007). The study of water infiltration is therefore crucial for its ecological significance.

Soil matrix and preferential flow are the two main forms of water infiltration into the soil (Jarvis et al., 2012). Matrix flow refers to the relatively slow movement of water and solutes through the soil, while passing through all pore spaces (Allaire et al., 2009). Preferential flow is defined as the physical movement of water and solutes along certain pathways, while bypassing a fraction of the pore matrix (Hendrickx and Flury, 2001). Preferential flow may accelerate or delay the movement of water and solutes depending upon the position of the water and solutes compared to the position of the preferential flow paths (Allaire et al., 2009). However, preferential flow is randomly distributed in the

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soil, spatially and temporally, and thus is essentially unpredictable (Hendrickx and Flury, 2001; Zhang et al., 2015, 2017). Preferential flow occurs in most soils and is caused from a series of factors such as macropore flow, non-homogeneous infiltration, lateral flow, and wetting front instabilities (Allaire et al., 2009; Bundt et al., 2001).

To visualize preferential water flow pathways in soils, dye traces are often used (Flury and Flühler, 1994; Ghodrati and Jury, 1990; Flury and Wai, 2003). Brilliant Blue FCF (C. I. Food Blue 2) has often been used for its lower adsorption and higher mobility (Flury and Flühler, 1994), although ions and isotopes have been also employed to distinguish preferential and matrix flow paths (e.g., Bogner et al., 2008; Backnäs et al., 2012; Ma et al., 2017; Jaromir and Tomas, 2018). Zhang et al. (2015) used Brilliant Blue solution, at a concentration of  $5 \text{ g L}^{-1}$ , to identify preferential flow pathways through dye coverage; and the maximum depth of infiltration could also be determined. Weiler and Flühler (2004), and Cey and Rudolph (2009) suggested that the type of macropore flow and the interaction between macropore flow and matrix flow can be understood after mapping dye concentration by using classification techniques or image analysis software.

Numerous methods for measuring infiltration have been developed and applied, such as the single ring infiltrometer (Reynolds and Elrick, 1990; Wu et al., 1999), the double-ring infiltrometer (Bouwer, 1986), the disc permeameter (Perroux and White, 1988; Ankeny et al., 1988; López-Vicente and Álvarez, 2018), the rainfall simulator (Peterson and Bubbenzer, 1986; Ogden et al., 1997; Mao et al., 2011), the point source method (Mao et al., 2016), the run off-on-out method (Liu et al., 2010), and the Guelph permeameter (López-Vicente and Navas, 2009; Rodrigo Comino et al., 2016, 2017). The ring infiltrometer is the most commonly used method for measuring soil infiltration rates (Walsh and McDonnell, 2012). According to Zhang et al. (2016), and using the single-ring infiltrometer, preferential flow encourages higher initial infiltration rates. Wu et al. (1997) compared the constant and falling head methods with numerical modelling, obtaining similar soil infiltration rates in fine textured soils, although soil infiltration rates decreased more than 30% with the decrease of the water head in coarse soils. The double-ring infiltrometer is frequently used to measure infiltration because this device is relatively inexpensive, simple and easy to operate; nevertheless, errors in the experimental results are not infrequent and are well documented. Zhang et al. (2016), for example, described the effect of preferential flow on the measurement accuracy of infiltration rates caused by the disturbance of soil when infiltrometer is inserted into the ground. Zhang et al. (2017), in order to avoid preferential flow in measuring infiltration in forest soil, added a layer of fine sand to the inner and outer rings and found that the measured infiltration reflected the transient process of soil matrix infiltration and the results were consistent with the values generated by accepted infiltration models. The disc permeameter, unlike the infiltrometer, forces water to penetrate into the soil under negative water potentials previously determined by the operator; therefore, it can only measure the hydraulic characteristics of unsaturated soil (Logsdon and Jaynes, 1993; Angulo-Jaramillo et al., 2000). Hence, this method cannot measure soil infiltration on soil surface under positive pressure conditions. Besides, the soil hydraulic conductivity measurement under ponded infiltration conditions is not ensured by using this method (Reynolds and Zebchuk, 1996). Rainfall simulation is often used to measure infiltration rates (Smets et al., 2011; Iserloh et al., 2013), and study soil erosion and other soil hydrological processes (Lassu et al., 2015; Keesstra et al., 2016). It is known that the different methods and techniques provide distinct soil water infiltrations rates under the same conditions. Cerda (1997) reported that the infiltration rates estimated by the ring infiltrometer were 8–8.5 times higher than those obtained by rainfall simulator based on 40 groups of field measurements. Furthermore, Mao et al. (2011) revealed that the initial infiltration rate measured by the rainfall simulator was not accurate under heavy rainfall conditions.

A point source method was developed by Su (2007) for measuring in

situ infiltration rates, obtaining lower rates than those rates measured with the linear source method; although the point source method had higher accuracy and lower relative error. The use of this method formed a circular or near-circular wet area at the soil surface, which increased gradually over time to reach a final wet area (Mao et al., 2016). Channels formed by plant roots are common preferential soil flow pathways both for water and solutes (Bogner et al., 2010; Germann et al., 2012; Zhang et al., 2017). The alfalfa's root system is characterized by a large-diameter, long, almost-straight taproot. When alfalfa dies and decays, it provides a temporary channel for macropore flow (Mitchell et al., 1995). Old root channels that have been opened at the soil surface (Wang and Strong, 1996), and root channels created by decayed roots in the vertical direction (Mitchell, 1995), are both regarded as macropores. Water infiltration into soil through root channels is an important mechanism of soil infiltration and plays an important role in predicting runoff generation and groundwater recharge (Weiler and Naef, 2003). Macropores formed by plant roots are crucial for the soil water downward movement, especially in arid and semi-arid areas. Wu et al. (2017) found that the initial infiltration rate and the average infiltration rate during the first 15 min increased as the diameter of root channels increased. Mitchell (1995), and Benegas et al. (2014) concluded that tree roots, including both living and decaying roots, have positive effects on soil infiltration by increasing macroporosity and soil aggregation. More than sixty years ago, Barley (1954) also illustrated that plant roots increased the soil infiltration rate.

Although previous studies have addressed the effects of root systems on soil water infiltration, the influence of root channels formed by decayed roots is scarcely studied. This theme, especially in arid and semi-arid regions, where soil-water availability for plant is essential, deserves to be studied in depth. The objectives of this study were: (1) to determine the decayed roots in response to preferential flow paths, and (2) to further explore the relationships between the properties of the channels formed by the decayed roots, and the soil water infiltration process.

## 2. Materials and methods

### 2.1. Study sites

This study was conducted at the Key Field Observation Station of Ecological Environment of the Ministry of Agriculture on the Loess Plateau, which is located in Gongjiawan district, Lanzhou city, in Gansu Province, China ( $36^{\circ}01'N$ ,  $103^{\circ}45'W$ ), as shown in Fig. 1. This area belongs to the arid hilly and gully regions on the Loess Plateau (altitude ranging from 1698 to 1823 m a.s.l.), which is characterized by a semi-arid and continental climate. Between 2001 and 2013, the mean annual precipitation and evaporation were 383 and 1450 mm, respectively. Rainfall events were unevenly distributed, and mainly concentrated in July, August and September. The average annual temperature was  $7.4^{\circ}\text{C}$ , and the highest and lowest recorded temperature were  $19.8^{\circ}\text{C}$  in July and  $-7.2^{\circ}\text{C}$  in January. The mean annual sunshine duration was about 2651 h, and the frost-free period ranged between 135 and 167 days.

The soil is mainly loessial, and the loess layer thickness is approximately 150 m. Within the fine soil fraction ( $< 2 \text{ mm}$  of diameter), the mean content of sand, silt and clay is 25.5%, 66.1% and 8.4%, respectively (determined by laser diffraction using a Mastersizer 2000, Malvern Instruments), according to the USDA texture classification. The soil moisture at field capacity, by mass, ranges between 19.3% and 22.6% (Huang et al., 2017), and the mean soil organic matter content is ca.  $27.2 \text{ g kg}^{-1}$  for the 0–30 cm depth layer.

### 2.2. Experimental plots and infiltration measurements

The experimental site was a cultivated alfalfa (*Medicago sativa* L.) field, where plants remained dead for 3–5 years. Other grass species like

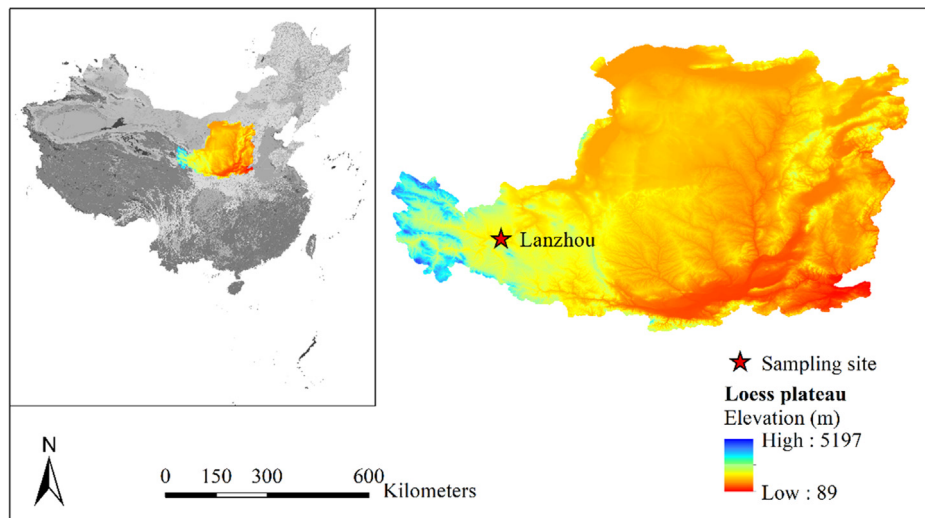


Fig. 1. Location of the experimental site on the Loess Plateau, Central China.

*Leymus secalinus*, *Artemisia sacrorum*, *Artemisia annua* and *Artemisia incisa*, etc. grew in the alfalfa field after the death of *Medicago sativa*. Three alfalfa field research areas, each area of  $10 \times 10$  m, were established in the grassland. Seeds were planted with a row spacing of 50 cm, and at a sowing rate of  $0.07 \text{ kg m}^{-2}$ . In the process of planting, the sowing rate was slightly higher than usual in order to improve the emergence rate, owing to the barren soil conditions.

Alfalfa grass growth depended on rainfall; there were no additional fertilizations, human interventions or animal disturbances, and these conditions were similar in all plots during all the experimental treatments. These treatments included alfalfa grassland and bare land; in addition, the alfalfa grassland was treated with or without a fine sand layer. The purpose of the fine sand layer was to avoid the occurrence of preferential flow. Six experimental plots were randomly selected at each experimental treatment and set as six replicates. Infiltration measurements were carried out in all the treatments: alfalfa, with (6 measurements) and without (6 measurements) fine sand, and bare soil (6 measurements). The double-ring infiltrometer was used to measure water infiltration (Fig. 2).

After selecting a level site, the litter layer was carefully removed so that the soil surface was exposed. Then, the rings (30-cm diameter inner ring, 60-cm diameter outer ring, both 20-cm in height) were placed concentrically on the soil surface and were gently inserted 10 cm vertically into the soil, in order to reduce crack formation. Some soil was then placed outside of the inner and outer rings to avoid water outflow. The inner and outer rings were filled with water, with and without

methylene blue, respectively, until the water depth reached 5 cm at the same time in both rings. When methylene blue was dissolved in water, the solution appeared bright blue. The falling head method was used to measure water infiltration. The time required to infiltrate water, up to the water level dropped 1 cm in the inner ring, was recorded with a stopwatch until the infiltration time did not change for three consecutive measurements, and this rate was assumed as the steady-state infiltration. During the infiltration tests, water was refilled into the rings whenever the water level dropped to 1 cm depth, and the water levels in the inner and outer rings were consistent. To calculate the initial infiltration rate, only the first three minutes of the infiltration process were taken into consideration. In fact, the initial infiltration rate was an average rate because we took into account the total three minutes of infiltrated water. The average of the last three fairly constant infiltration rates was considered as the steady-state infiltration rate. These rates were calculated following the equation of Zhang et al. (2017):

$$i = \Delta I / \Delta t \times 600 \quad (1)$$

where  $i$  is the infiltration rate ( $\text{mm h}^{-1}$ ),  $\Delta I$  is the cumulative infiltration in the inner ring (cm) during  $\Delta t$  time of infiltration (min), and 600 is a unit conversion factor.

To evaluate the role played by the root channels in the preferential flow, other double-ring infiltrometer was applied to measure the infiltration process in the alfalfa grassland with the fine sand layer (6 measurements). According to the methods of Zhang et al. (2017), a

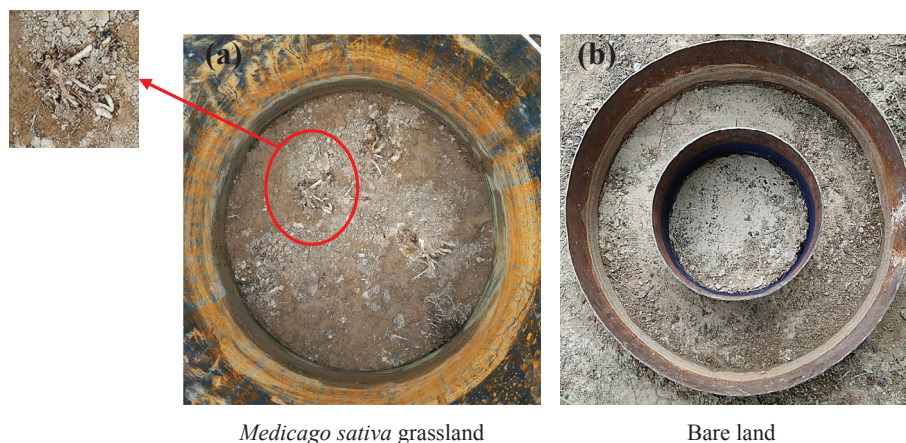


Fig. 2. Pictures of the infiltrometer rings used to measure the soil water infiltration in two experimental treatments.



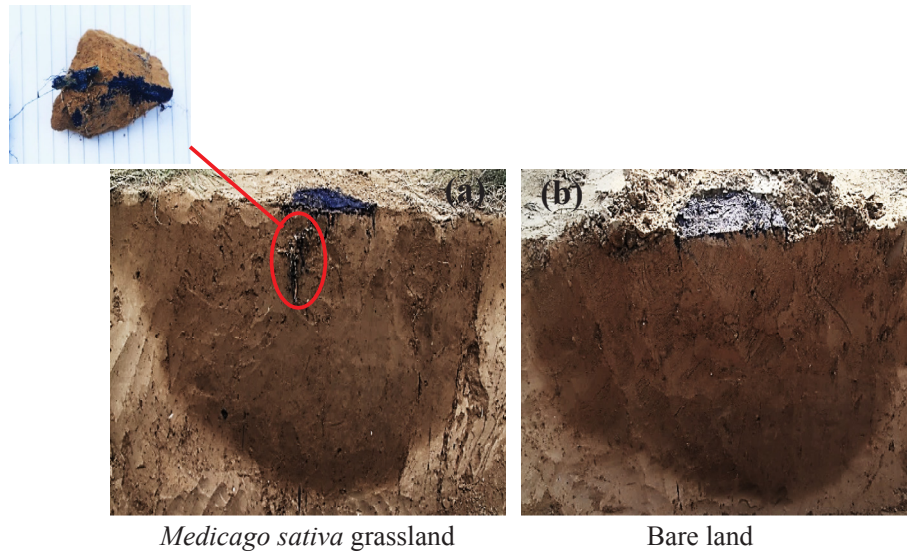


Fig. 3. Pictures of the vertical soil profiles below the inner infiltrometer rings 24 h after ending the infiltration tests, in two experimental treatments.

layer of nylon cloth was placed on the soil surface inside the inner ring and between the inner and outer rings, and then a layer of fine sand, of 1–3 cm thick, was evenly placed on the nylon cloth, in order to avoid the occurrence of preferential flow and to ensure that water infiltrates the soil surface only through the soil matrix. To reduce the scouring of the fine sand layers while the rings were filled with water, paperboards were placed on the sand surface. In the infiltration tests with sand, measurements were carried out following the same method mentioned above.

### 2.3. Soil sampling and analysis

Twenty-four hours after the end of the infiltration tests, vertical soil profiles were excavated passing through the center of the inner ring. The vertical soil profiles are shown in Fig. 3 for alfalfa and bare land treatments. The width and depth of the wetted soil area were then measured with a steel ruler, having the smallest division of 1 mm. The wetted vertical profiles were recorded with a camera, in order to distinguish the spots stained by the tracer dye. Afterwards, soil samples were collected throughout the entire soil profile (0–100 cm), at 10-cm intervals, on the other side of the excavation trench. Three replicates were collected at each soil layer, and the soil water content by mass (SWC, % weight) and soil bulk density (BD,  $\text{g cm}^{-3}$ ) were determined. The SWC was measured by taking the proportion of the loss of mass, after oven-drying at  $105^\circ\text{C}$ , to the constant mass of dry soil. The BD was measured using a stainless-steel cutting ring sampler of 5-cm diameter and height ( $393 \text{ cm}^3$ ) to collect the samples. The soil total porosity (TP, %) was calculated using the following equation:

$$TP = (1 - BD/ds) \times 100 \quad (2)$$

where  $ds$  is the soil particle density ( $\text{g cm}^{-3}$ ), which was assumed to be  $2.65 \text{ g cm}^{-3}$  (Huang et al., 2017). The mean and standard error values of the soil physical properties are shown in Table 1.

### 2.4. Root channels measurement

The diameter of the root channels, i.e. the diameter of the stubbles on the ground, were only measured in the inner ring of the twelve alfalfa experimental plots by using a Vernier caliper (having a Vernier constant of 0.10 mm). The average value of the stubble diameters was regarded as the average diameter of root channels, which we assumed that have influenced the preferential water flow (Wu et al., 2017). Meanwhile, to estimate the area of the root channels, the total number

Table 1

Mean  $\pm$  standard error values of the soil physical properties (SWC: soil water content by mass; BD: soil bulk density; TP: soil total porosity) for the entire soil profile (0–100 cm).

| Treatments | Soil depth (cm) | SWC (%)          | BD ( $\text{g cm}^{-3}$ ) | TP (%)            |
|------------|-----------------|------------------|---------------------------|-------------------|
| Grassland  | 0–10            | $2.67 \pm 0.02a$ | $1.31 \pm 0.03a$          | $50.10 \pm 1.24a$ |
|            | 10–20           | $3.87 \pm 0.86a$ | $1.33 \pm 0.58a$          | $49.98 \pm 1.94a$ |
|            | 20–30           | $3.69 \pm 0.53a$ | $1.48 \pm 0.05a$          | $44.15 \pm 2.04a$ |
|            | 30–40           | $4.19 \pm 1.26a$ | $1.38 \pm 0.07a$          | $47.75 \pm 2.69a$ |
|            | 40–50           | $4.07 \pm 0.92a$ | $1.23 \pm 0.03b$          | $53.42 \pm 1.28a$ |
|            | 50–60           | $3.56 \pm 0.52a$ | $1.24 \pm 0.07b$          | $53.07 \pm 2.79a$ |
|            | 60–70           | $3.58 \pm 0.63a$ | $1.23 \pm 0.06a$          | $53.60 \pm 2.42a$ |
|            | 70–80           | $3.58 \pm 0.02b$ | $1.22 \pm 0.01b$          | $53.86 \pm 0.41a$ |
|            | 80–90           | $3.39 \pm 0.24a$ | $1.24 \pm 0.03b$          | $53.08 \pm 1.26a$ |
|            | 90–100          | $3.14 \pm 0.11a$ | $1.24 \pm 0.01b$          | $53.07 \pm 0.57a$ |
| Bare land  | 0–10            | $2.71 \pm 0.03a$ | $1.28 \pm 0.07a$          | $51.57 \pm 2.51a$ |
|            | 10–20           | $3.91 \pm 0.87a$ | $1.36 \pm 0.09a$          | $48.49 \pm 3.28a$ |
|            | 20–30           | $3.72 \pm 0.58a$ | $1.49 \pm 0.04a$          | $43.78 \pm 1.45a$ |
|            | 30–40           | $4.26 \pm 1.31a$ | $1.48 \pm 0.02a$          | $43.94 \pm 0.65a$ |
|            | 40–50           | $4.14 \pm 0.12a$ | $1.45 \pm 0.06a$          | $45.22 \pm 2.23b$ |
|            | 50–60           | $3.67 \pm 0.58a$ | $1.44 \pm 0.05a$          | $45.49 \pm 1.81b$ |
|            | 60–70           | $3.69 \pm 0.71a$ | $1.32 \pm 0.03a$          | $50.04 \pm 1.21a$ |
|            | 70–80           | $3.70 \pm 0.14a$ | $1.36 \pm 0.04a$          | $48.71 \pm 1.63b$ |
|            | 80–90           | $3.48 \pm 0.33a$ | $1.35 \pm 0.03a$          | $49.06 \pm 2.16b$ |
|            | 90–100          | $3.21 \pm 0.18a$ | $1.33 \pm 0.02a$          | $49.81 \pm 1.84a$ |

Note: Real density was assumed to be  $2.65 \text{ g cm}^{-3}$ . For each soil depth but different treatment, the values followed by a different letter are significantly different at 0.05 level.

of root channels was counted in the inner ring at each experimental site. According to the methods of Wu et al. (2017), the following formulas were used to calculate the average root channels diameter (ARCD, cm), and the root channel area (RCA,  $\text{cm}^2$ ).

$$ARCD = \sum_{i=1}^n d/n \quad (n = 1, 2, 3, \dots) \quad (3)$$

$$RCA = n \cdot \pi \cdot ARCD^2 / 4 \quad (n = 1, 2, 3, \dots) \quad (4)$$

where  $d_i$  is the diameter of the root channel (cm),  $n$  is the number of root channels.

### 2.5. Statistical analysis

The statistical analyses were conducted using SPSS software (ver. 22.0). One-way analysis of variance (ANOVA) was used to determine if

the differences in the soil properties between each experimental treatment were statistically significant with a p-value of 0.05. Regression analysis was used to determine relationships between the maximum infiltration depth, the diameter of root channels, the area of root channels, the number of root channels and the steady-state infiltration rate. The coefficient of determination ( $R^2$ ) was used to evaluate the performance of the applied regression equations.

### 3. Results

#### 3.1. Soil properties in different treatments

The differences in the soil physical properties of the different treatments are shown in Table 1. Overall, the SWC and BD were lower in the *Medicago sativa* grassland than in the bare land, whereas opposite differences were found for the TP. A significant difference between *Medicago sativa* grassland and bare soil was found for SWC for the 70–80 cm soil depth, although the differences were not significant for the other soil layers. The BD was significantly lower in *Medicago sativa* grassland than in bare land for all the soil depths greater than 40–50 cm. The TP was significantly higher in the *Medicago sativa* grassland than in the bare land for the soil depths of 40–50, 50–60, 70–80, and 80–90 cm.

#### 3.2. Soil water infiltration in the decayed root channels

The infiltration rates and cumulative infiltration curves of all the experimental treatments are shown in Fig. 4. The initial infiltration rate was high, as usual in this study, in all treatments; it decreased with time and finally reached a stable state. The different soil water infiltration rates in *Medicago sativa* were always higher than those rates found in the bare land (Fig. 4). Compared with the bare land, where the initial infiltration rate was  $190.48 \text{ mm h}^{-1}$ , the same rate in the *Medicago sativa* grassland was 27.70% greater, reaching up to  $243.24 \text{ mm h}^{-1}$  (Figs. 4a, 5). At the end of the experiment, the cumulative infiltration in the alfalfa fields was 1.13 times the value obtained in the bare land (Fig. 4b). In addition, our results revealed significant differences in the steady-state infiltration rate between the different experimental treatments (Fig. 5). The steady-state infiltration rate in the *Medicago sativa* grassland was higher than that observed in the bare land. In particular, the steady-state infiltration rate in the alfalfa grassland ( $73.32 \text{ mm h}^{-1}$ ) was 31.80% greater than the value estimated in the bare land ( $55.62 \text{ mm h}^{-1}$ ). Moreover, when the infiltration rate reached a steady-

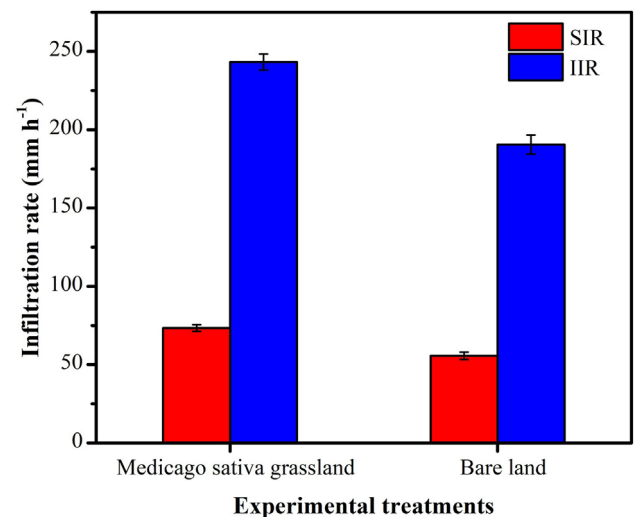


Fig. 5. Mean infiltration rates and their standard deviation in two different experimental treatments. Note: SIR is the steady-state infiltration rate; IIR is the initial infiltration rate (first initial 15 min for each test).

state value, there was a positive and strong correlation ( $R^2 = 0.825$ ) between the steady-state infiltration rate and the maximum depth reached by the water along the soil profile (Fig. 6). This means that the higher depth the water penetrated into the soil, the higher steady-state infiltration rates were reached. The maximum depth reached by the infiltrated water was ca. 96 cm in the alfalfa grassland, and ca. 63 cm in the bare land plots (Fig. 3). We concluded that preferential flowpaths increased water penetration into the soil, owing to the decay of the alfalfa roots.

#### 3.3. Relation between root channel properties and soil water infiltration

The correlation analyses showed that the average root channel diameter was positively correlated ( $R^2 = 0.815$ ,  $p < 0.01$ ) with the steady-state infiltration rate (Fig. 7b). Likewise, there was a significant relationship ( $R^2 = 0.789$ ,  $p < 0.01$ ) between the root channel area and the steady-state infiltration rate (Fig. 7c). In other words, the infiltration rate increased with the increase of the ARCD and RCA values. The number of root channels was also significantly and positively correlated ( $R^2 = 0.797$ ,  $p < 0.01$ ) with the steady-state infiltration rate (Fig. 7a).

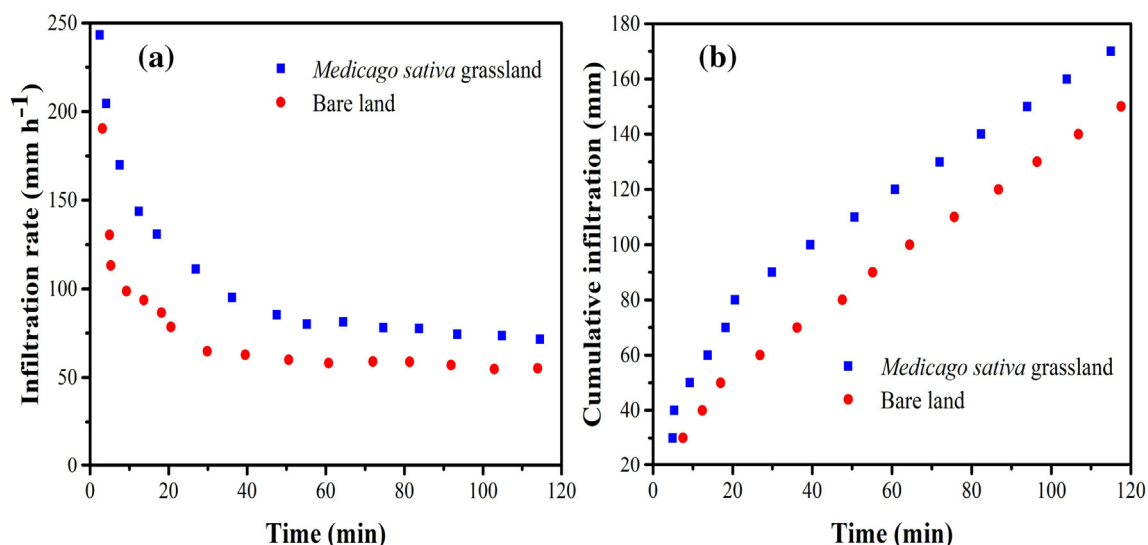


Fig. 4. Infiltration rate (a) and cumulative infiltration (b) curves for the experimental treatments.

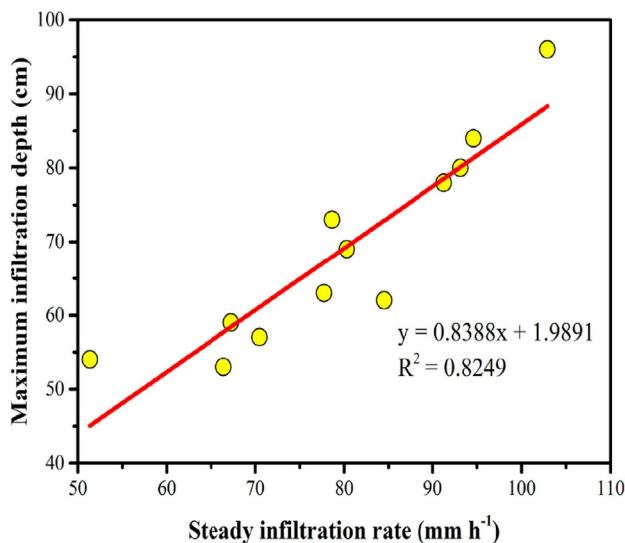


Fig. 6. Relationship between the steady-state infiltration rate and the soil water infiltration depth. Note: The maximum infiltration depth, indicated on the vertical axis, is the maximum depth the water reached along the soil profile after 24 h.

#### 4. Discussion

Soil water infiltration is related to surface runoff, soil erosion, plant water storage and groundwater recharge (Jiang et al., 2018). In addition, water flow behavior is increasingly considered as the main factor of soil water distribution (Jiang et al., 2017). In our study, the steady-state infiltration rate of alfalfa was higher than that of the bare land. Consistent with the finding of Meek et al. (1990), the effect of alfalfa-root-induced macropores was mainly demonstrated by the increase of final infiltration rate when alfalfa roots decayed. Schematically, there are two main forms of water infiltration in the soil: matrix flow and preferential flow (Jarvis et al., 2012). Soil matrix flow, which is present in unsaturated conditions, is affected by capillary suction, and the process of infiltration is well matched with Philip's sorptivity concept (Zhang et al., 2017). Preferential flow, mainly due to gravitational potential, which is generally in the vertical direction (Germann et al., 2007), is caused by a number of factors such as macropore flow, non-homogeneous infiltration, lateral flow and wetting front instabilities (Allaire et al., 2009; Bundt et al., 2001) and, therefore, is difficult to

describe with current theories, such as Richard's and Darcy-type equations (De Vries and Chow, 1978; Weiler, 2005).

Brilliant Blue dye is an important tool to prove the existence of preferential flow (Flury et al., 1994). Our study showed that most of the dye covered the top soil (Fig. 3) and this result was in agreement with the findings of Devitt and Smith (2002). Several studies showed that the dye coverage was lower in the subsoil than at the soil surface, because preferential flow was more frequent in the upper soil layers than in the deeper ones (Zhang et al., 2015). Furthermore, Mitchell et al. (1995) illustrated that the dyeing depth went below 55 cm along the decayed root channels, more likely because the roots influenced the physical properties of the soil, such as porosity and bulk density (Aubertin, 1971). The decayed root channels, tagged with blue dye inside (Fig. 3a), were a clear indication of pore space creation by the alfalfa roots. Dye was found around the roots, and the dye coverage decreased with the soil depth. When a root decayed, a clear porous soil remained, which was characterized by lower density and higher porosity, providing new space for water and air storage.

Our results were in agreement with those of Mitchell et al. (1995) and Barley (1954), who indicated that root channels, created by decayed roots, increased the infiltration rate. The existence of preferential flow resulted in the enhancement of soil infiltration capacity. After alfalfa died, the channels formed by the decayed roots were the main preferential flow that provided a temporary channel for macropore flow (Mitchell et al., 1995), as shown in Fig. 3a. Root channels would increase macropores network density and continuity, and channels formed by plant roots may be conducive to water transport and solute movement as preferential flow (Bogner et al., 2010; Germann et al., 2012). In general, and compared with living roots, it was much easier for decayed roots to form long and continuous preferential flow pathways that facilitated the downward movement of water (Mitchell et al., 1995). Additionally, we also found that the ARCD and RCA showed a positive correlation with the infiltration rates (Fig. 7b, c). This result agreed with the observations reported by Wu et al. (2017), who illustrated that the ARCD and RCA were positively correlated with the average infiltration rate calculated during the 0–15 min time interval and the initial infiltration rate. Lastly, it must be noted that the roots of legumes, such as alfalfa, are more effective creating stable macropores than those of gramineous plants, which contribute to increasing infiltration rates and benefit the higher proliferation and weight of leguminous roots (Wu et al., 2016).

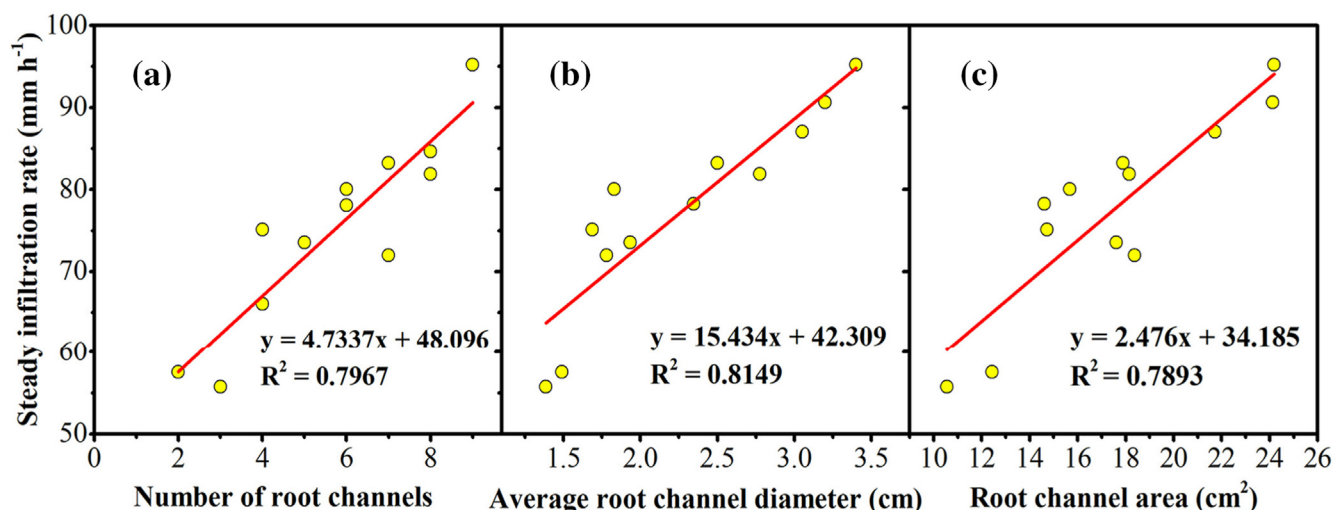


Fig. 7. Relationships between the number of root channels (a), the average root channel diameter (b) and the root channel area (c) vs. the steady-state infiltration rate.



## 5. Conclusions

This study demonstrated that the decayed root channels of alfalfa had significant effects on soil water infiltration in a loessial cultivated soil under semi-arid conditions. Compared to bare soil, the initial soil water infiltration rate in the *Medicago sativa* grassland increased by 27.70% and the cumulative infiltration was 1.13 times higher. Moreover, the steady infiltration rate in the grassland increased by 31.80%, compared with the bare land control plots. Our results indicated that root channels created by the alfalfa decayed roots played a significant role in increasing infiltration rate. The root diameter and area of the root sections, indicated previously with the terms ARCD and RCA, were significantly and positively correlated with the infiltration rate, being respectively 0.815 and 0.789 the coefficients of determination. Our results supported the conclusion that the decayed root channels of alfalfa significantly enhanced infiltration of water from soil surface toward deeper layers. Further research on this phenomenon could contribute to better understanding on the land surface water hydrological cycle process in arid areas.

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